# GEMM — Complexity Decomposition Tree

Each level multiplies its factor → product = total cost

**Algorithm 6: Main loop**

for each stable degree $s = 0 \dots D$

$D + 1$ iterations

$D = \left\lfloor \frac{R-n}{p-1} \right\rfloor$

**Algorithms 3–4: Recursion tree**

branching $\beta$, depth $\leq \log_p s$

$N(s) \leq \beta^{\log_p s}$ leaves

$\sum_s N(s) = S = O(D^{1+\log_p \beta})$

**Algorithm 5: Künneth product**

$H \leftarrow H \otimes_K H_*(E_k)$ for each sequence

1 per sequence

**Output degree loop**

for $m = 0, 1, \dots, R$

$R + 1$ iterations

**Sparse inner loop**

for $j \in \mathrm{supp}(E_k)$

$|\mathrm{supp}(E_k)| = R/\delta_k$ pairs

$\delta_k = n + (p-1)\, s_k$ (Lemma 7.6)

**$\otimes$ and Tor on Group pairs**

bilinear on cyclic components

$O(C^2)$ per pair

$C = f + \lfloor \log_p R \rfloor + 1$ (Cor. 7.4)

$$\underbrace{\sum_{s=0}^{D} \frac{N(s)}{\delta(s)}}_{\text{levels } 0\text{–}1\text{–}4} = O\left(\frac{D^{\log_p \beta}}{p-1}\right)$$

$$\text{Total} = \underbrace{\frac{D^{\log_p \beta}}{p-1}}_{\text{sequences}/\delta} \times \underbrace{R}_{\text{degrees}} \times \underbrace{C^2}_{\text{pairs}} = \frac{R^{2+\log_p \beta}}{(p-1)^{1+\log_p \beta}} \cdot (f + \log_p R)^2$$

For fixed $p, f, n$: $\boxed{O\left(R^{2+\log_p \beta} \cdot \log^2 R\right)}$ — Theorem 7.5 (sparse Künneth)